Introduction: This worksheet is designed to give you more practice with setting up and computing Riemann sums In practice, more complicated approximations are usually done with the help of computer algebra systems. You are strongly encouraged to use a calculator.

Part I: Setting Up Riemann Sum – Partitions and Sketching Rectangles

Problem 1 Before approximating the area under the curve

$$f(x) = \frac{4}{49}x^2 + 1 \text{ on the interval } [0,7] \qquad a=0,\ b=7$$

using $n = 7$ equally sized subintervals, determine the following:

(a) The step size Δx.

(b) All partition points and subintervals.

(c) Sketch the rectangles corresponding to the **Left Endpoint Riemann Sum.**

(d) Sketch the rectangles corresponding to the **Right Endpoint Riemann Sum.**

(e) Sketch the rectangles corresponding to the **Midpoint Riemann Sum.**

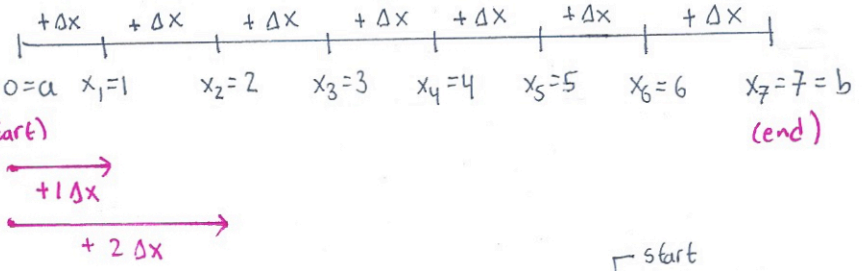
These sketches represent the regions whose areas we will approximate in problem 2.

Solution

(a) **Stepsize** (equally sized subintervals $\Rightarrow$ even partition)

$$\Delta x = \frac{b-a}{n} = \frac{7-0}{7} = 1.$$

(b) **Partition Points and Subintervals**

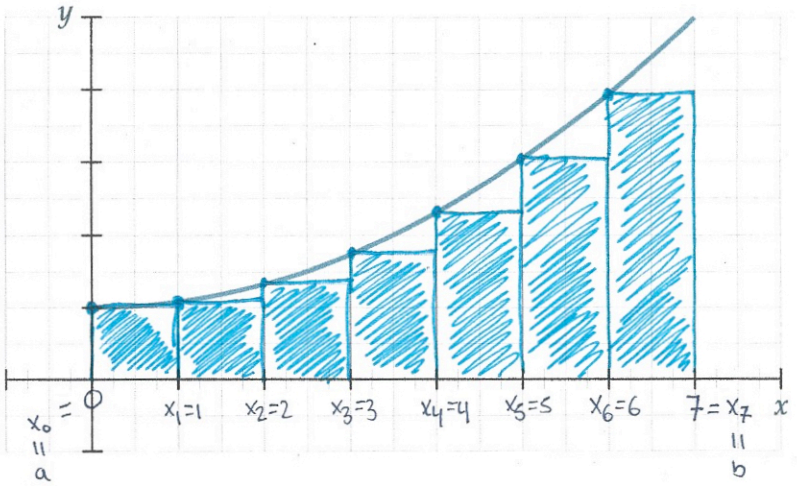


Sequentially find partition points using: $x_k = a + k\Delta x$ for $k = 0,1,2,\dots,7$. (start; move k steps of size Δx)

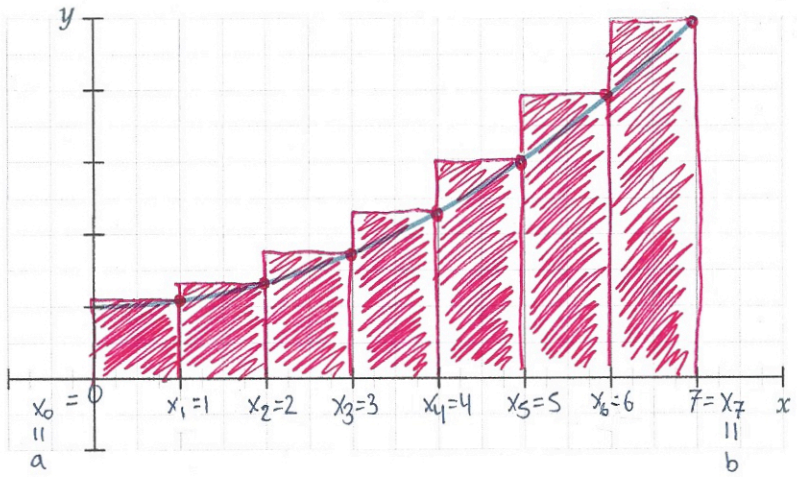
$\Rightarrow$ Partition points: $\{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\} = \{0,1,2,3,4,5,6,7\}$ in the x-direc.

$\Rightarrow$ Subintervals: $[0,1]$, $[1,2]$, $[2,3]$, $[3,4]$, $[4,5]$, $[5,6]$, $[6,7]$.

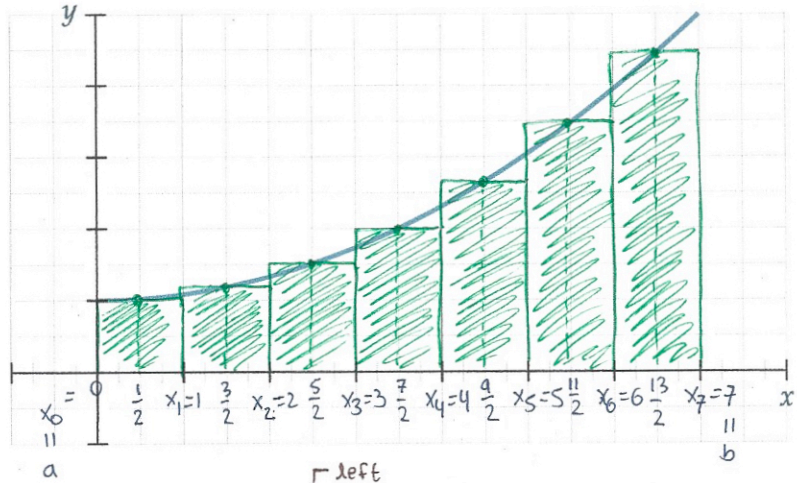
(c) **Left Endpoint Riemann Sum with $n = 7$:**



(d) **Right Endpoint Riemann Sum with $n = 7$:**



(e) **Midpoint Riemann Sum with $n = 7$:**



midpoints : $x_k^m = \dfrac{x_k + x_{k+1}}{2}$ (left, right) for $k = 0, 1, 2, \ldots, 6$.

□

Part II : Computing Riemann Sum–Approximations of Area

Problem 2 Approximate the area under the curve

$$f(x) = \frac{4}{49}x^2 + 1 \text{ on the interval } [0,7]$$

using $n = 7$ equally-sized subintervals.

(a) Compute the **Left Endpoint Riemann Sum.**

(b) Compute the **Right Endpoint Riemann Sum.**

(c) Compute the **Midpoint Riemann Sum.**

(d) The **exact** area under the graph of f is $\frac{49}{3} \approx 16.33333$. List your three Riemann sum approximations in order from smallest to largest, and indicate which one is closest to the exact area.

Solution

(a) **Left Endpoint Riemann Sum**

The left endpoints are : $x_0 = 0,\ x_1 = 1,\ x_2 = 2,\ x_3 = 3,\ x_4 = 4,\ x_5 = 5,\ x_6 = 6$.

Thus $$L_7 = \sum_{k=0}^{6} f(x_k)\,\Delta x$$

$$= f(x_1)\,\Delta x + f(x_2)\,\Delta x + f(x_3)\,\Delta x + f(x_4)\,\Delta x + f(x_5)\,\Delta x + f(x_6)\,\Delta x$$

* Note : when doing Riemann sum with even partition, factoring out Δx makes algebra easier.

$$= \Delta x \left[\sum_{k=1}^{6} f(x_k) \right]$$

$$= 1 + \frac{53}{49} + \frac{65}{49} + \frac{85}{49} + \frac{113}{49} + \frac{149}{49} + \frac{193}{49}$$

$$= \frac{707}{49}$$

$$\Rightarrow L_7 \approx 14.43.$$

(b) Right Endpoint Riemann Sum

The right endpoints are: $x_1 = 1$, $x_2 = 2$, $x_3 = 3$, $x_4 = 4$, $x_5 = 5$, $x_6 = 6$, $x_7 = 7$

Thus $$R_7 = \sum_{k=1}^{7} f(x_k)\,\Delta x$$

$$= f(x_1)\underbrace{\Delta x}_{L_1} + f(x_2)\underbrace{\Delta x}_{=1} + f(x_3)\underbrace{\Delta x}_{=1} + f(x_4)\underbrace{\Delta x}_{=1} + f(x_5)\underbrace{\Delta x}_{=1} + f(x_6)\underbrace{\Delta x}_{=1} + f(x_7)\underbrace{\Delta x}_{=1}$$

$$= \frac{53}{49} + \frac{65}{49} + \frac{85}{49} + \frac{113}{49} + \frac{149}{49} + \frac{193}{49} + 5$$

$$= \frac{903}{49}$$

$$\Rightarrow R_7 \simeq 18.43.$$

(c) Midpoint Riemann Sum

The midpoints are: $x_0^m = \frac{1}{2}$, $x_1^m = \frac{3}{2}$, $x_2^m = \frac{5}{2}$, $x_3^m = \frac{7}{2}$, $x_4^m = \frac{9}{2}$, $x_5^m = \frac{11}{2}$, $x_6^m = \frac{13}{2}$.

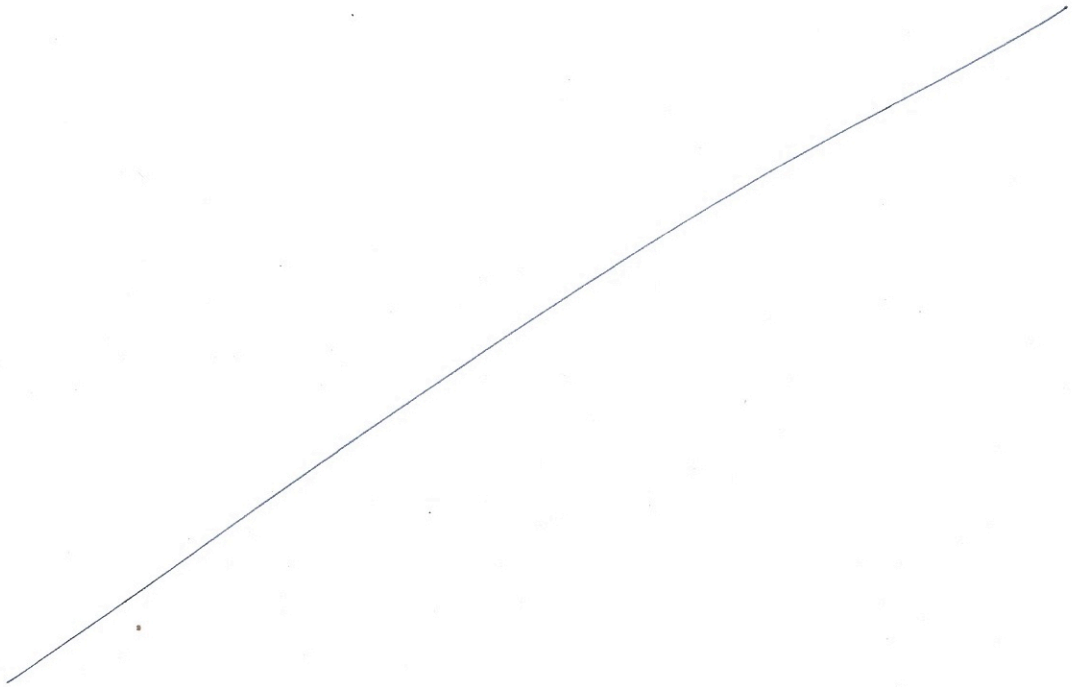
Thus $$M_7 = \sum_{k=0}^{6} f(x_k^m)\,\Delta x$$

$$= f(x_0^m)\underbrace{\Delta x}_{=1} + f(x_1^m)\underbrace{\Delta x}_{=1} + f(x_2^m)\underbrace{\Delta x}_{=1} + f(x_3^m)\underbrace{\Delta x}_{=1} + f(x_4^m)\underbrace{\Delta x}_{=1} + f(x_5^m)\underbrace{\Delta x}_{=1} + f(x_6^m)\underbrace{\Delta x}_{=1}$$

$$= \frac{50}{49} + \frac{58}{49} + \frac{74}{49} + \frac{98}{49} + \frac{130}{49} + \frac{170}{49} + \frac{218}{49}$$

$$= \frac{798}{49}$$

$$\Rightarrow M_7 \simeq 16.29.$$



(d) Comparing Results

Exact area under the curve : $\frac{49}{3} \approx 16.3\overline{3}$

we have $L_7 \approx 14.43 < M_7 \approx 16.29 < R_7 \approx 18.43$

$\Rightarrow$ midpoint Riemann sum was the closest.

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